

Maurya P Gowda

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PROFILE

Information Science & Engineering undergraduate at BMS College of Engineering with experience in building backend systems, data-driven applications, and machine learning projects. Interested in software engineering, applied machine learning, and system design. Strong focus on learning through implementation and real-world problem solving.

COMPETENCIES

- **Programming & Development:** Python, C, C++, Java (fundamentals), backend application development using Flask
- **Databases:** MySQL, MongoDB
- **Machine Learning:** Supervised & unsupervised learning models, feature extraction and model evaluation, *Libraries:* scikit-learn, NumPy, Pandas, Librosa, PyTorch
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code
- **Core Computer Science Concepts:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Cryptography, Network Security

EDUCATION

BMS College of Engineering, Basavangudi, Bengaluru
Bachelor of Engineering in Information Science (6th Semester)

2023 - ongoing
9.14 CGPA

PROJECTS

Smart Meta-Controller for AI Routing

- Built an intelligent routing system using a DistilBERT classifier to dynamically assign user queries to lightweight or deep AI models based on task complexity.
- Implemented a two-stage architecture with proactive classification and reactive escalation for fault tolerance and reliable task handling.
- Evaluated system performance on large workloads and measured computational efficiency and response latency.
- **Impact:** Achieved **62% reduction in energy consumption**, **52% reduction in response latency**, and **68% reduction in computational load**, contributing towards sustainable and green AI deployment.
- **Technologies:** Python, PyTorch, DistilBERT, Machine Learning

Kannada Music Genre Classification System

- Developed a machine learning pipeline to classify Kannada music into Dance, Folk, Devotional, and Romantic genres using audio feature extraction.
- Curated a regional dataset and extracted MFCC, spectral, and rhythm-based features using Librosa.
- Trained and evaluated multiple supervised models including SVM, Random Forest, Gradient Boosting, and Voting Ensemble.
- **Impact:** Achieved **87.5% classification accuracy** using ensemble learning, supporting digital organization and preservation of regional Indian music.
- **Technologies:** Python, scikit-learn, Librosa

IoT Data Dashboard

- Developed backend services to collect and process real-time sensor data from IoT devices.
- Implemented **APIs** for data ingestion and device monitoring.
- Built interactive dashboards for live **visualization** and **analytics**.
- **Impact:** Enables **real-time monitoring** of distributed devices, supporting **predictive maintenance** and **data-driven decision making**.
- **Technologies:** Python, Flask, Data Processing Libraries

Other Projects

- **HALO-CME Prediction using Aditya-L1 Data** - Machine learning model to predict Earth-directed solar events using Aditya-L1 solar wind data.
- **Diabetes Prediction System** - ML-based health risk prediction using patient medical data.
- **Hotel Reservation & Management System** - Backend-driven hotel booking and management platform with user authentication and database integration.

PUBLICATIONS

Smart Meta-Controller: Energy Efficient Dynamic Routing for LLM Inference Systems - <https://ieeexplore.ieee.org/document/11295492>

EXTRACURRICULAR

- **Chairperson**, BMSCE IEEE Power & Energy Society and Sensors Council (Organize technical hackathons, workshops, seminars etc.)
- **Performance Tracking team member**, IEEE PES SYP Global Mentoring Initiative
- **IEEE PES DAY Ambassador**, Global IEEE PES DAY Celebration
- **Senior Coordinator** for Stage & Stalls, PhaseShift (Annual College Tech Fest)